Related work

1. PID control and its extensions

The PID controller is the most used controller in industry [6], [8], [10], and the articles about canals say the same [5]. The first rules for tuning it come from Ziegler and Nichols [9], and in practice the weights are often found by trial. We also found our weights by trial, with a search in the program (chapter 8 in modeling).

- **Anti-windup [7], [11].** When the valve is at its limit, the integral can grow without effect, and this is called windup. The usual solution is to stop adding to the integral at the limit. Our PID+ does this. Our controller limits the integral to ±1500 and it also clears it when the error changes sign. This is different from the usual anti-windup, and in our tests it helps at the start but it hurts when the null position u_0 is not 50.
- **Filtering and noise [13].** Filtered PID and PIDD2 controllers for integrating processes with measurement noise are described in [13]. A tank level is an integrating process. Our smoothing (the average with the previous value) has the same purpose, and our tests show that it is the most important part of our controller.
- **Second derivative (PIDD2) [12], [13], [14].** These articles give tuning rules for controllers with a second derivative, for oscillatory systems with delay [12], for integrating processes [13] and for processes with inertia and delay [14]. The rules start from a model of the process. We use the second derivative too ($\frac{3}{5}$ of the rate and $\frac{2}{5}$ of the acceleration in one derivative part), but we found the weights by search, without a model.